

MoBI Laboratory In Practice: Examples from two studies

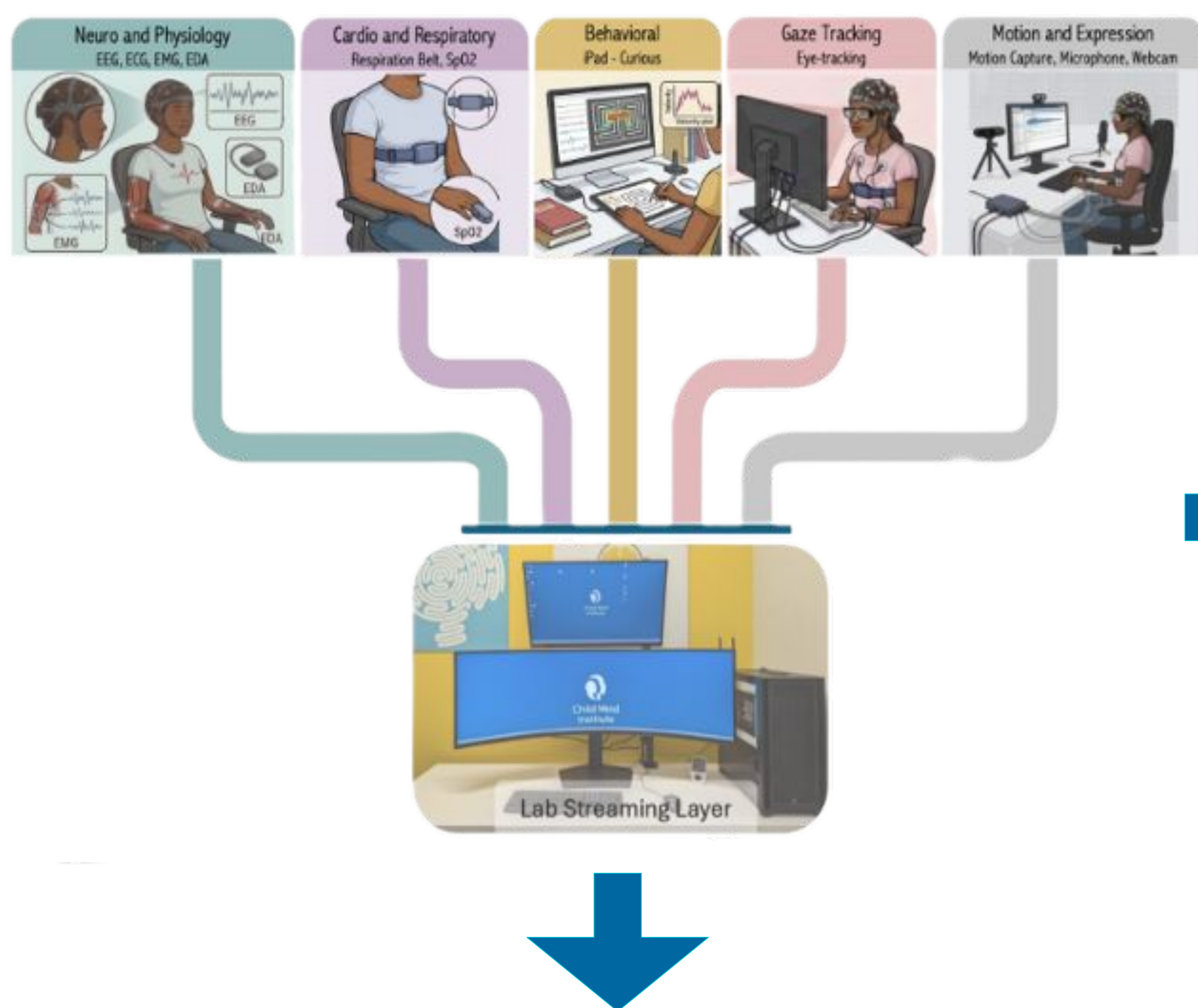
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Introduction

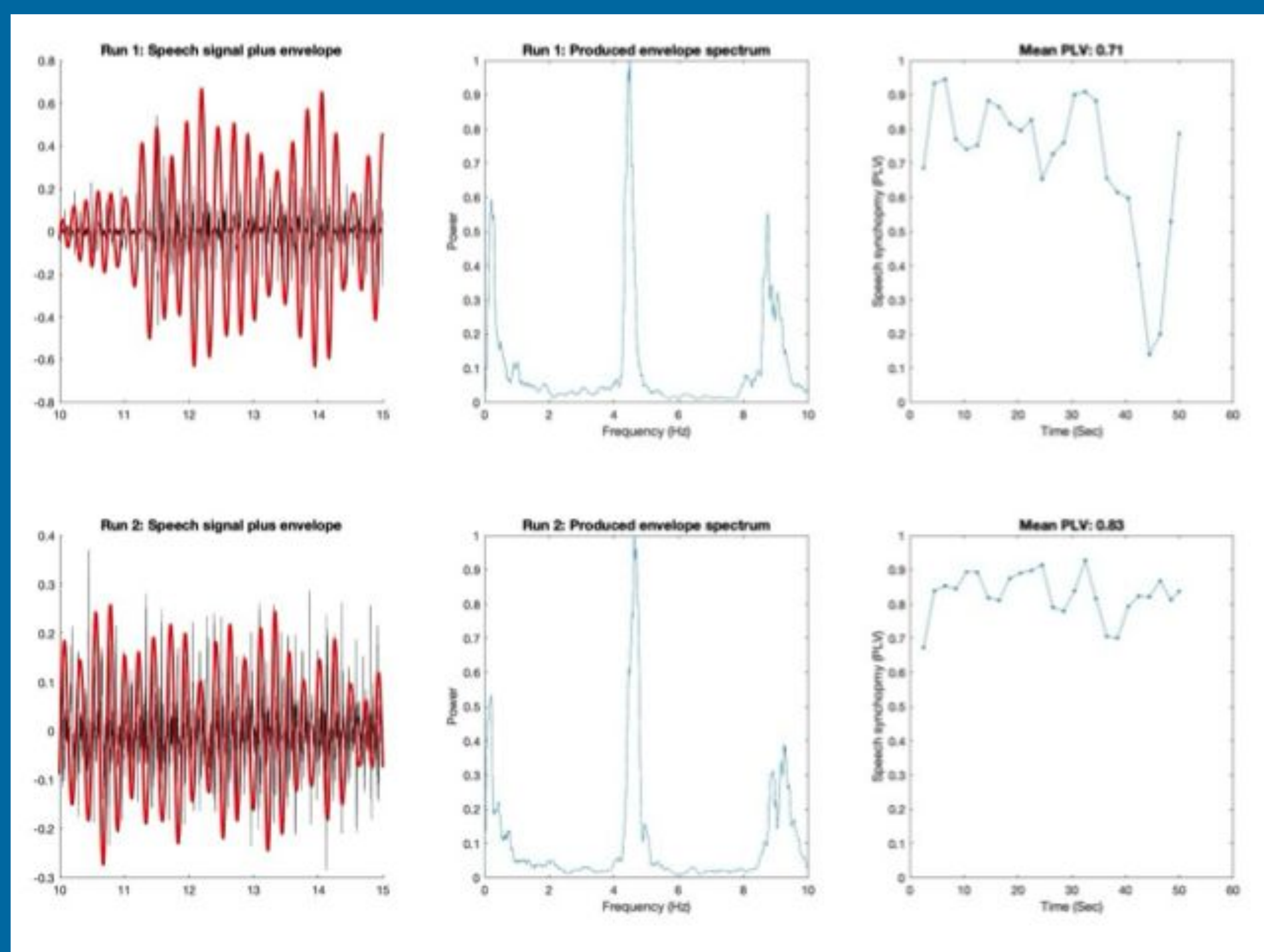
- Multimodal brain/body imaging (MoBI) laboratories offer unique opportunities to capture the complex interplay between neural, physiological, and behavioral processes during data collection.^{1,2}
- Building upon our previous work establishing a single-computer system that utilizes Lab Streaming Layer (LSL), we present preliminary findings from two ongoing studies that demonstrate the laboratory's capabilities for multimodal data collection: **Graphomotor Control** and **Auditory-Motor Synchronization**.

Method and Preliminary Results

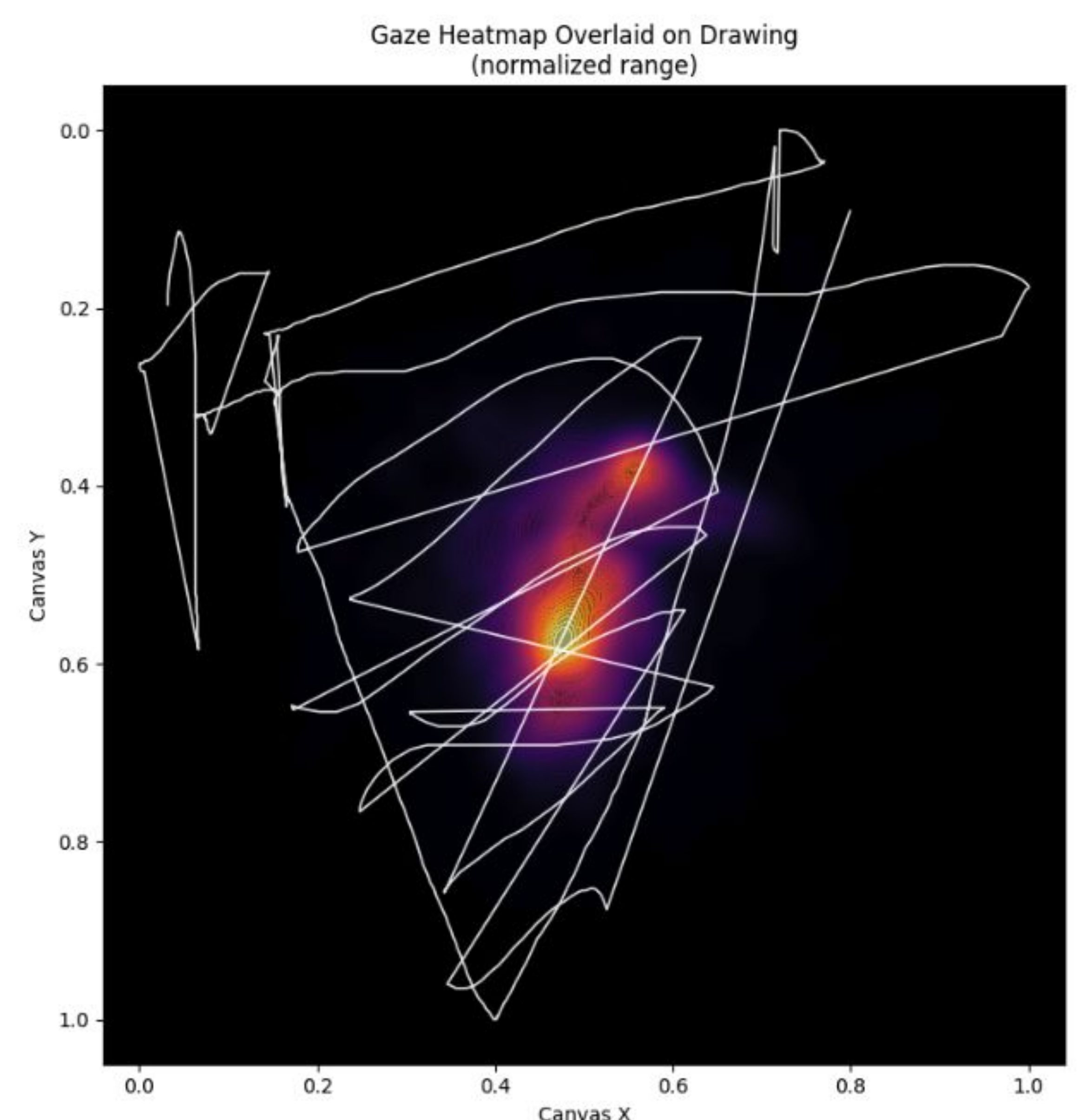
MoBI Lab with LSL synchronized data



Auditory-Motor Synchronization



Graphomotor Control



The above graph shows both **graphomotor drawing data** and **eye-tracking data** that was captured **at the same time** by our system. All data is temporally synchronized, which says time needing to align data post-hoc.

Discussion

- The synchronized multimodal recordings provide rich, complementary data that would be difficult to obtain using traditional single-modality approaches.
- Our team has built 5 MoBI labs to date and data collection is ongoing in all 5!

SCAN ME !



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References:

- [1] Makeig et al., 2009
- [2] Gramann et al., 2011